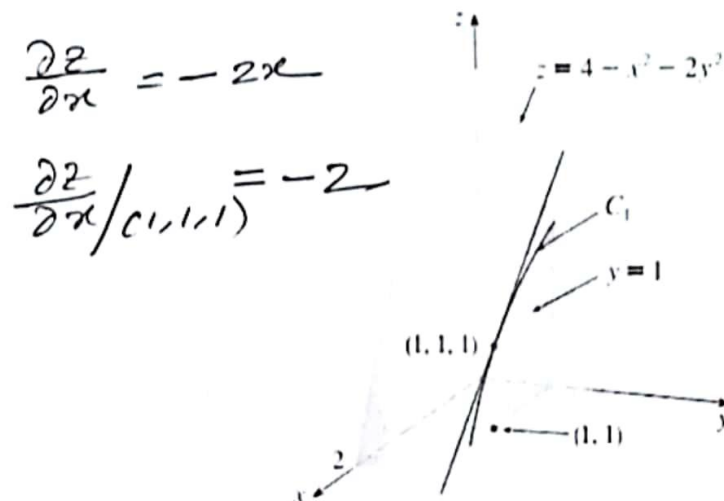


Math-2183 Section-B Class test-02

1. Find the slope of the tangent line indicating in the following figure. [2]



The function is decreasing in the indicated direction.

2. Show that $u(x, t) = \sin(kx) \sin(mkt)$ is a solution of the one dimensional wave equation $u_{tt} = m^2 u_{xx}$. [3]
3. For $e^{-xy} + \frac{1}{z} = 2$ evaluate $\frac{\partial z}{\partial x}$ by the implicit and explicit approaches. Show that those are the same. [3]
4. For $u = \tan(x^2 - y^2)$ show that $\frac{1}{x} \frac{\partial u}{\partial x} + \frac{1}{y} \frac{\partial u}{\partial y} = 0$. [2]

$$u_t = \sin(kx) \cos(mkt) mk$$

$$u_{tt} = -\sin(kx) \sin(mkt) m^2 k$$

$$u_x = \cos(kx) \sin(mkt) k$$

$$u_{xx} = -\sin(kx) \sin(mkt) k^2$$

SO, $u_{tt} = m^2 u_{xx}$

$$f(x, y, z) = e^{-xy} + \frac{1}{z} - 2$$

$$\frac{\partial f}{\partial x} = -y e^{-xy}$$

$$\frac{\partial f}{\partial z} = -\frac{1}{z^2}$$

$$\frac{\partial z}{\partial x} = -\frac{\partial f / \partial x}{\partial f / \partial z} = -\frac{-y e^{-xy}}{-1/z^2} = -y z^2 e^{-xy}$$

$$e^{-xy} + \frac{1}{z} = 2$$

$$\Rightarrow -y e^{-xy} - \frac{1}{z^2} \frac{\partial z}{\partial x} = 0$$

$$\Rightarrow \frac{\partial z}{\partial x} = -y z^2 e^{-xy}$$

$$u = \tan(x^2 - y^2)$$

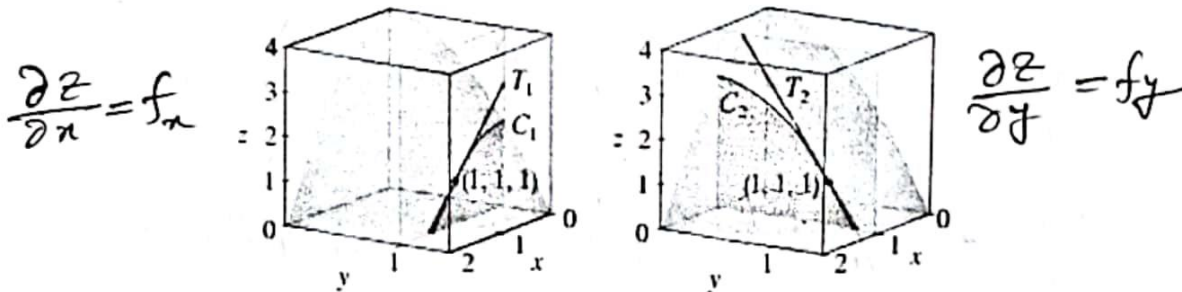
$$\frac{\partial u}{\partial x} = \sec^2(x^2 - y^2) 2x$$

$$\frac{\partial u}{\partial y} = \sec^2(x^2 - y^2) (-2y)$$

$$\Rightarrow \frac{1}{x} \frac{\partial u}{\partial x} + \frac{1}{y} \frac{\partial u}{\partial y} = 2 \sec^2(x^2 - y^2) - 2 \sec^2(x^2 - y^2) = 0$$

Math-2183 Section-C Class test-02

1. How could you find the slopes of the following tangent lines for $z = f(x, y)$? [1]



2. The temperature at a point (x, y) on a flat metal plate is given by $T(x, y) = \frac{60}{1+x^2+y^2}$. Find the rate of change of temperature in y -direction at the point $(-2, 1)$. [2.5]
3. Determine whether $u(x, y) = \sin x \cosh y + \cos x \sinh y$ satisfies the two dimensional Laplace equation $u_{xx} + u_{yy} = 0$. [2.5]
4. Using the partial differentiation find $\frac{dy}{dx}$ for the function $\ln(xy) = 1$. [1.5]
5. For $w = \ln(x^3 + y^3 + z^3)$ show that $x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} + z \frac{\partial w}{\partial z} = 3$. [2.5]

$$\frac{\partial T}{\partial y} = \frac{-60}{(1+x^2+y^2)^2} (2y) = \frac{-120y}{(1+x^2+y^2)^2}$$

$$\therefore \frac{\partial T}{\partial y} \bigg|_{(-2, 1)} = \frac{-120}{36} < 0, \text{ temperature decreasing.}$$

$$u_x = \cos x \cosh y - \sin x \sinh y$$

$$u_{xx} = -\sin x \cosh y - \cos x \sinh y$$

$$u_y = \sin x \sinh y + \cos x \cosh y$$

$$u_{yy} = \sin x \cosh y + \cos x \sinh y$$

$$\text{So, } u_{xx} + u_{yy} = 0.$$

$$f(x, y) = \ln(xy) - 1$$

$$\frac{\partial f}{\partial x} = \frac{1}{xy} y = \frac{1}{x}$$

$$\frac{\partial f}{\partial y} = \frac{1}{xy} x = \frac{1}{y}$$

$$\frac{dy}{dx} = - \frac{\partial f / \partial x}{\partial f / \partial y} = - \frac{1/x}{1/y} = - \frac{y}{x}$$

$$* \text{ check: } \ln(xy) = 1 \Rightarrow \frac{1}{xy} (y + x \frac{dy}{dx}) = 0$$

$$\Rightarrow y + x \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = - \frac{y}{x}$$

$$\textcircled{5} \quad w = \ln(x^3 + y^3 + z^3) = \ln(u) \quad ; \quad u = x^3 + y^3 + z^3$$

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial x} = \frac{1}{u} 3x^2 = \frac{3x^2}{x^3 + y^3 + z^3}$$

$$\frac{\partial w}{\partial y} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial y} = \frac{1}{u} 3y^2 = \frac{3y^2}{x^3 + y^3 + z^3}$$

$$\frac{\partial w}{\partial z} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial z} = \frac{1}{u} 3z^2 = \frac{3z^2}{x^3 + y^3 + z^3}$$

$$\therefore x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} + z \frac{\partial w}{\partial z} = \frac{3(x^3 + y^3 + z^3)}{x^3 + y^3 + z^3} = 3$$